NEWCOMER CREDIT DECISIONING COPILOT

Deciding, and defending, thin-file credit for UAE newcomers

A B2B copilot that gives a UAE credit officer a fast, consistent, defensible **approve, decline, or refer** decision on an applicant with no local credit history.

PMCurve AI-PM Capstone, Cohort 6 · Solo, bring-your-own project · Synthetic data only

THE PROBLEM

A new arrival starts with a blank credit file

- **The market:** the UAE is about 88% expat. A large share of loan applicants are recent arrivals.
- **The gap:** a newcomer has no record at the Al Etihad Credit Bureau (AECB), so the usual credit score does not exist.
- **Today:** banks either reject good customers outright, or assess them slowly and inconsistently by hand, the better part of an hour per file.
- **The cost:** lost good customers, slow onboarding, and decisions an officer cannot easily defend to a regulator.

88% expat is a market figure. All applicant data in this deck is synthetic.

WHO IT IS FOR

The credit officer, who has to stand behind the call

- **User:** a credit officer at a UAE digital bank, assessing newcomer personal-loan and credit-card applications.
- **Their job:** decide repay-vs-default risk, apply lending policy, and write a rationale they can defend later.
- **Their constraint:** in a regulated decision, being able to explain and reconstruct the call matters as much as the call itself.
- **The design rule:** the tool recommends; the human always makes the final decision.

THE PRODUCT

Structured intake to a defensible recommendation

1

Intake

Officer enters employment, salary, tenure, rent history, and a few alt-data fields.

2

Scorecard

A transparent 5-factor judgmental scorecard scores repay-vs-default risk.

3

Policy check

Six UAE lending rules run against the applicant, each with a citation.

4

Explanation

An LLM writes a plain-language rationale grounded only in the above.

5

Recommendation

Approve, decline, or refer. The officer reviews and makes the final call.

The score and the verdict are fully deterministic. The LLM writes the explanation only; it cannot change points, rules, or the recommendation.

THE APPROACH

Rules plus an LLM, with no trained model

- **What we built:** a rules-based scorecard and policy engine for the decision, with an LLM for the explanation layer only.
- **What we did not build:** a trained credit model. No black-box score drives the verdict.
- **Why:** for a regulated v1 lending decision, explainability and defensibility beat a marginal accuracy gain from a model nobody can audit.
- **The featured layer:** the explanation and decision-reasoning, the part a bank cannot get from a raw prompt.

A REAL DECISION

What the officer sees, end to end

INPUT

Government employee, 14 months in the UAE

Salary AED 20,000 / month, no obligations

On-time rent, 6+ months

Wants a personal loan: AED 40,000 over 12 months

OUTPUT: APPROVE · low risk · score 98/100

✓ Salary over double the AED 8,000 minimum; debt burden 18% (cap 50%)

✓ Government employer, top stability tier

✓ All six policy rules pass, each cited

✓ 6+ months on-time rent: strongest behavioral signal

A weaker profile returns **refer** for a human to check, or **decline** with the exact rule that failed named. Real output, captured from the running app.

An explanation you can trust, by construction

- **The use case:** the LLM turns the deterministic score and policy results into a plain-language rationale for the officer.
- **Grounding validator:** a check catches any invented fact, retries once, and falls back to a deterministic template, so no ungrounded text reaches the officer.
- **Result:** hallucination is enforced to zero, not just measured. In the eval, 24/24 explanations were grounded on the first attempt.
- **Adversarial probe:** a built-in prompt-injection test proves a malicious input in the name field cannot move the decision.

EVALUATION, THE TEST SET

An exam the model could not study for

- **24 locked profiles:** synthetic applicants spanning every rule and boundary, 8 approve, 8 decline, 8 refer.
- **Labels locked first:** outcomes were fixed before any scorecard weight existed. The labels could not be tuned to fit the model.
- **Method:** run all 24 through the full pipeline including the real model explanation, verify each against ground truth, compute metrics vs the Phase 1 targets.
- **Honesty rule:** boundary mismatches are surfaced, not tuned away. A mismatch is information about a borderline case, not a bug to hide.

EVALUATION, THE RESULTS

Every safety target met; one latency tail to fix

Metric	Target	Measured	Met
Decision accuracy	80% or higher	20/24 (83.3%)	Yes
False approval rate	under 10%	0/24 (0.0%)	Yes
Hallucination (shown)	0 across the set	0/24	Yes
Policy grounding	100% traceable	100%	Yes
Latency per assessment	under 15s	avg 8.2s, max 15.3s	Tail

Creditworthy newcomers approved: 7 of 8. The four accuracy misses are conservative refers or a cautious call; none is a false approval. The only miss vs target is the latency tail (one slow case at 15.3s).

WHY A BANK CANNOT JUST PROMPT AN LLM

The hard part is standing behind the decision

- **A prompt proves the model works once.** Production needs three things a prompt does not give you, and this build carries the demo-scale version of each:
- **Evals:** the 24-profile locked ground truth, run by a self-check harness.
- **Override tracking:** a review queue and audit log where a referred case and the analyst's action survive a reload.
- **Decision reconstruction:** a policy citation on every rule outcome and a ruleset version stamped on every decision, so you can answer 'why did we decline this person in June' months later.

THE ARCHITECTURE BET

Policy is configuration, a market is a pack

- **The rules live in a config pack,** not in code: rule values, citations, score tiers, and band cut-offs are all data.
- **Live what-if:** drag the debt-burden cap from 50% to 45% and the 24-profile run recomputes on stage, with no code change.
- **The shift is visible:** approvals drop from 7 of 8 to 6 of 8, two boundary cases move to decline, and every citation now reads 45%.
- **The payoff:** a second market (a new bureau and regulator) is a new pack, not a new project.

THE IMPACT

From the better part of an hour to minutes

TODAY · manual review

- Assemble salary, visa, obligations, bureau status by hand
- Check each against policy from memory or a PDF
- Write up the rationale
- ≈ the better part of an hour per file

WITH THE COPILOT

- Structured intake
- Policy rules run instantly, with citations
- Officer reviews recommendation and counterfactuals
- Judgment spent only on the refer queue, minutes

Illustrative, showing where the time goes; not a measured timing claim. A measured study is parked for v2.

WHAT IS NEXT

Parked on purpose, ready to answer 'and then?'

- **A second market pack:** Saudi Arabia, swapping AECB/CBUAE for SIMAH/SAMA. The real test of 'a market is a pack'. (Rule values illustrative until SAMA-sourced.)
- **Overrides as feedback:** every queue override is labeled data on where the rules disagree with human judgment, feeding rule tuning.
- **Version vs version:** a second ruleset (uae v1.1) and a version selector, so impact analysis becomes champion vs challenger.
- **Close the latency tail:** the one item still short of target, isolate the slow case and cap it.

Built, evaluated, and defensible

- All four capstone phases complete: scope, design, build, and evaluation.
- The MVP is built, type-clean, tested, and measured against a locked baseline.
- Every safety target met (0 false approvals, 0 hallucinations, 100% grounding).
- Live deployment is the one remaining step.

github.com/Mxjoshi/newcomer-credit-copilot

Product, scope, and sign-offs: Monika.   Engineering pair: Claude Code.